

Performance Testing

6.Project Testing | Credit Card Approval Prediction

Prepared For	Prepared By	Domain	Date
Credit Card Approval Prediction	Ankitha Vana	Finance	July 1, 2026

Testing Objective

Testing verifies that the selected model predicts reliably, the Flask application accepts valid inputs, and the prediction response is fast enough for real-time use.

Model Evaluation Metrics

Metric	Purpose
Accuracy	Measures overall percentage of correct predictions.
Precision	Measures how many predicted approvals/rejections are correct.
Recall	Measures how many actual risky or eligible cases are detected.
F1-score	Balances precision and recall for imbalanced classes.
Confusion Matrix	Shows true positives, true negatives, false positives, and false negatives.

Performance Test Plan

Test Case	Expected Result
Valid applicant profile	Application returns Approved or Rejected prediction.
Missing required field	Form validation prevents incomplete submission or shows an error.
High-risk credit status	Model classifies applicant as likely rejected when overdue indicators are strong.
Repeated prediction requests	Application remains responsive under repeated submissions.
Cloud scoring endpoint	IBM Watson ML endpoint returns prediction in expected response format.

Model Comparison Template

After running the training notebook, record the measured values in this table and select the highest-performing model for deployment.

Model	Accuracy	Precision	Recall	F1-score	Selected?
Logistic Regression	Record after run	Record after run	Record after run	Record after run	No/Yes
Decision Tree	Record after run	Record after run	Record after run	Record after run	No/Yes
Random Forest	Record after run	Record after run	Record after run	Record after run	No/Yes
XGBoost	Record after run	Record after run	Record after run	Record after run	No/Yes